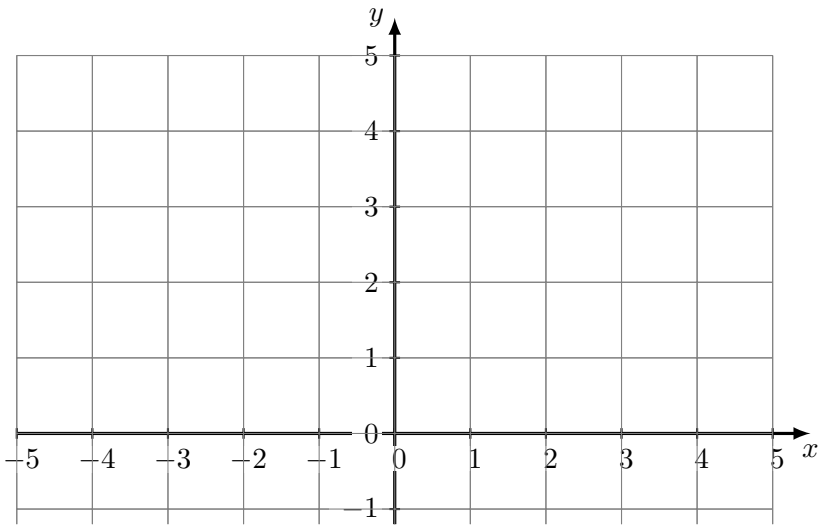
1. Sketch the function $f(x) = |x| + |1 + x|$ on the grid below by adding together the two absolute value functions.



2. Find the piecewise function description for $f(x) = |x| + |1 + x|$ using the definition of absolute value, which is reprinted below

$$|x| = \begin{cases} x & \text{if } x \geqslant 0 \\ -x & \text{if } x < 0 \end{cases}$$

3. Use Mathematica's **Piecewise[]** function to graph what you found in part **C.**